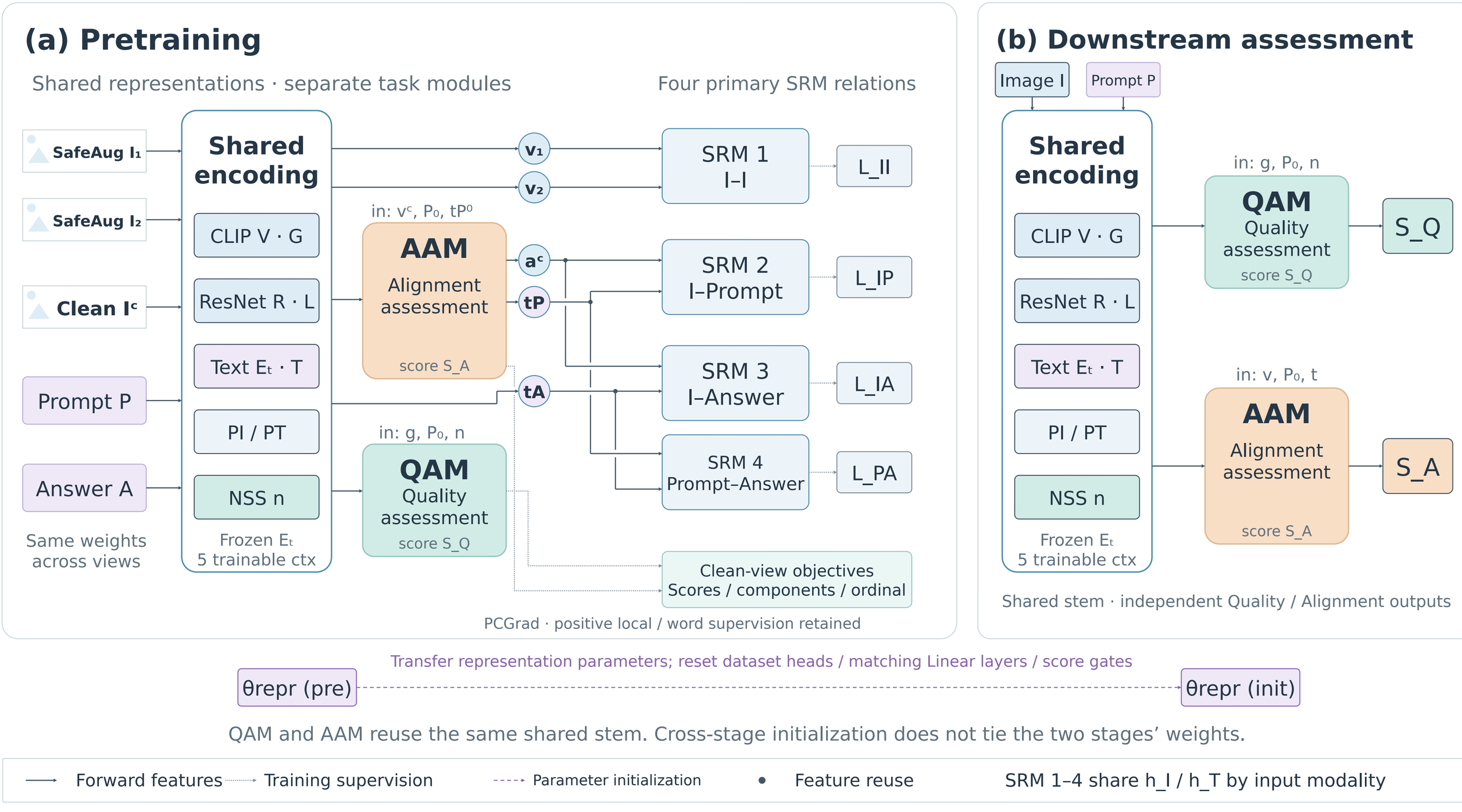

(a) Pretraining
Shared representations · separate task modules
Four primary SRM relations
SafeAug I₁
SafeAug I₂
Clean I^c
Prompt P
Answer A
Same weights across views
Shared encoding
CLIP V · G
ResNet R · L
Text Et · T
PI / PT
NSS n
Frozen Et
5 trainable ctx
v₁
v₂
in: v^c, P₀, tP⁰
AAM
Alignment assessment
score S_A
a^c
tP
tA
in: g, P₀, n
QAM
Quality assessment
score S_Q
SRM 1
I–I
SRM 2
I–Prompt
SRM 3
I–Answer
SRM 4
Prompt–Answer
L_II
L_IP
L_IA
L_PA
Clean-view objectives
Scores / components / ordinal
PCGrad · positive local / word supervision retained
(b) Downstream assessment
Image I
Prompt P
Shared encoding
CLIP V · G
ResNet R · L
Text Et · T
PI / PT
NSS n
Frozen Et
5 trainable ctx
in: g, P₀, n
QAM
Quality assessment
score S_Q
S_Q
in: v, P₀, t
AAM
Alignment assessment
score S_A
S_A
Shared stem · independent Quality / Alignment outputs
Transfer representation parameters; reset dataset heads / matching Linear layers / score gates
θrepr (pre)
θrepr (init)
QAM and AAM reuse the same shared stem. Cross-stage initialization does not tie the two stages' weights.
Forward features
Training supervision
Parameter initialization
Feature reuse
SRM 1–4 share h_I / h_T by input modality